

Lecture 17

Gradient

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Consider the three-dimensional scalar field $f = f(x, y, z)$, and the differential df , given by

$$df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy + \frac{\partial f}{\partial z} dz.$$

Using the dot product, we can write this in vector form as

$$df = \left(\frac{\partial f}{\partial x} \mathbf{i} + \frac{\partial f}{\partial y} \mathbf{j} + \frac{\partial f}{\partial z} \mathbf{k} \right) \cdot (dx \mathbf{i} + dy \mathbf{j} + dz \mathbf{k}) = \nabla f \cdot d\mathbf{r},$$

where in Cartesian coordinates,

$$\nabla f = \frac{\partial f}{\partial x} \mathbf{i} + \frac{\partial f}{\partial y} \mathbf{j} + \frac{\partial f}{\partial z} \mathbf{k}$$

is called the gradient of f . The nabla symbol ∇ is pronounced “del” and ∇f is pronounced “del- f ”. Another useful way to view the gradient is to consider ∇ as a vector differential operator which has the form

$$\nabla = \mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z}.$$

Because of the properties of the dot product, the differential df is maximum when the infinitesimal displacement vector $d\mathbf{r}$ is along the direction of the gradient ∇f . We then say that ∇f points in the direction of maximally increasing f , and the magnitude of ∇f gives the slope (or gradient) of f in that direction.

Example: Compute the gradient of $f(x, y, z) = xyz$.

The partial derivatives are easily calculated, and we have

$$\nabla f = yz \mathbf{i} + xz \mathbf{j} + xy \mathbf{k}.$$

Problems for Lecture 17

1. Let the position vector be given by $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, with $r = |\mathbf{r}| = \sqrt{x^2 + y^2 + z^2}$. Compute the gradient of the following scalar fields and write the result in terms of $\mathbf{r}$ and r .

a) $\phi(x, y, z) = x^2 + y^2 + z^2;$

b) $\phi(x, y, z) = \frac{1}{\sqrt{x^2 + y^2 + z^2}}$

Solutions to the Problems

Lecture 18

Divergence

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Consider in Cartesian coordinates the three-dimensional vector field, $\mathbf{u} = u_1(x, y, z)\mathbf{i} + u_2(x, y, z)\mathbf{j} + u_3(x, y, z)\mathbf{k}$. The divergence of $\mathbf{u}$, denoted as $\nabla \cdot \mathbf{u}$ and pronounced “del-dot-u”, is defined as the scalar field given by

$$\begin{aligned}\nabla \cdot \mathbf{u} &= \left(\mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z} \right) \cdot (u_1\mathbf{i} + u_2\mathbf{j} + u_3\mathbf{k}) \\ &= \frac{\partial u_1}{\partial x} + \frac{\partial u_2}{\partial y} + \frac{\partial u_3}{\partial z}.\end{aligned}$$

Here, the dot product is used between a vector differential operator ∇ and a vector field $\mathbf{u}$. The divergence measures how much a vector field spreads out, or diverges, from a point. A more math-based description will be given later.

Example: Let the position vector be given by $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$. Find $\nabla \cdot \mathbf{r}$.

A direct calculation gives

$$\nabla \cdot \mathbf{r} = \frac{\partial}{\partial x}x + \frac{\partial}{\partial y}y + \frac{\partial}{\partial z}z = 3.$$

Example: Let $\mathbf{F} = \frac{\mathbf{r}}{|\mathbf{r}|^3}$ for all $\mathbf{r} \neq 0$. Find $\nabla \cdot \mathbf{F}$.

Writing out the components of $\mathbf{F}$, we have

$$\mathbf{F} = F_1\mathbf{i} + F_2\mathbf{j} + F_3\mathbf{k} = \frac{x}{(x^2 + y^2 + z^2)^{3/2}}\mathbf{i} + \frac{y}{(x^2 + y^2 + z^2)^{3/2}}\mathbf{j} + \frac{z}{(x^2 + y^2 + z^2)^{3/2}}\mathbf{k}.$$

Using the quotient rule for the derivative, we have

$$\frac{\partial F_1}{\partial x} = \frac{(x^2 + y^2 + z^2)^{3/2} - 3x^2(x^2 + y^2 + z^2)^{1/2}}{(x^2 + y^2 + z^2)^3} = \frac{1}{|\mathbf{r}|^3} - \frac{3x^2}{|\mathbf{r}|^5},$$

and analogous results for $\partial F_2/\partial y$ and $\partial F_3/\partial z$. Adding the three derivatives results in

$$\nabla \cdot \mathbf{F} = \frac{3}{|\mathbf{r}|^3} - \frac{3(x^2 + y^2 + z^2)}{|\mathbf{r}|^5} = \frac{3}{|\mathbf{r}|^3} - \frac{3}{|\mathbf{r}|^3} = 0,$$

valid as long as $|\mathbf{r}| \neq 0$, where $\mathbf{F}$ diverges. In electrostatics, $\mathbf{F}$ is proportional to the electric field of a point charge located at the origin.

Problems for Lecture 18

1. Find the divergence of the following vector fields:

a) $\mathbf{F} = xy\mathbf{i} + yz\mathbf{j} + zx\mathbf{k};$

b) $\mathbf{F} = yz\mathbf{i} + xz\mathbf{j} + xy\mathbf{k}.$

Solutions to the Problems

Lecture 19

Curl

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Consider in Cartesian coordinates the three-dimensional vector field $\mathbf{u} = u_1(x, y, z)\mathbf{i} + u_2(x, y, z)\mathbf{j} + u_3(x, y, z)\mathbf{k}$. The curl of $\mathbf{u}$, denoted as $\nabla \times \mathbf{u}$ and pronounced “del-cross-u”, is defined as the vector field given by

$$\nabla \times \mathbf{u} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ u_1 & u_2 & u_3 \end{vmatrix} = \left(\frac{\partial u_3}{\partial y} - \frac{\partial u_2}{\partial z} \right) \mathbf{i} + \left(\frac{\partial u_1}{\partial z} - \frac{\partial u_3}{\partial x} \right) \mathbf{j} + \left(\frac{\partial u_2}{\partial x} - \frac{\partial u_1}{\partial y} \right) \mathbf{k}.$$

Here, the cross product is used between a vector differential operator and a vector field. The curl measures how much a vector field rotates, or curls, around a point. A more math-based description will be given later.

Example: Show that the curl of a gradient is zero, that is, $\nabla \times (\nabla f) = 0$.

We have

$$\begin{aligned} \nabla \times (\nabla f) &= \begin{pmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ \partial f/\partial x & \partial f/\partial y & \partial f/\partial z \end{pmatrix} \\ &= \left(\frac{\partial^2 f}{\partial y \partial z} - \frac{\partial^2 f}{\partial z \partial y} \right) \mathbf{i} + \left(\frac{\partial^2 f}{\partial z \partial x} - \frac{\partial^2 f}{\partial x \partial z} \right) \mathbf{j} + \left(\frac{\partial^2 f}{\partial x \partial y} - \frac{\partial^2 f}{\partial y \partial x} \right) \mathbf{k} = 0, \end{aligned}$$

using the equality of mixed partials.

Example: Show that the divergence of a curl is zero, that is, $\nabla \cdot (\nabla \times \mathbf{u}) = 0$.

We have

$$\begin{aligned} \nabla \cdot (\nabla \times \mathbf{u}) &= \frac{\partial}{\partial x} \left(\frac{\partial u_3}{\partial y} - \frac{\partial u_2}{\partial z} \right) + \frac{\partial}{\partial y} \left(\frac{\partial u_1}{\partial z} - \frac{\partial u_3}{\partial x} \right) + \frac{\partial}{\partial z} \left(\frac{\partial u_2}{\partial x} - \frac{\partial u_1}{\partial y} \right) \\ &= \left(\frac{\partial^2 u_1}{\partial y \partial z} - \frac{\partial^2 u_1}{\partial z \partial y} \right) + \left(\frac{\partial^2 u_2}{\partial z \partial x} - \frac{\partial^2 u_2}{\partial x \partial z} \right) + \left(\frac{\partial^2 u_3}{\partial x \partial y} - \frac{\partial^2 u_3}{\partial y \partial x} \right) = 0, \end{aligned}$$

again using the equality of mixed partials.

Problems for Lecture 19

1. Find the curl of the following vector fields:

a) $\mathbf{F} = xy\mathbf{i} + yz\mathbf{j} + zx\mathbf{k}$;

b) $\mathbf{F} = yz\mathbf{i} + xz\mathbf{j} + xy\mathbf{k}$.

2. Consider a two-dimensional velocity field given by

$$\mathbf{u} = u_1(x, y)\mathbf{i} + u_2(x, y)\mathbf{j}.$$

Show that the vorticity $\boldsymbol{\omega} = \nabla \times \mathbf{u}$ takes the form

$$\boldsymbol{\omega} = \omega_3(x, y)\mathbf{k}.$$

Determine ω_3 in terms of u_1 and u_2 .

Solutions to the Problems